Paper Title

First Author
Department
Institution
City, Country
first.last@example.com

Abstract

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

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1 Introduction

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.TODO!

The remainder of the paper starts with a presentation of related work (Section 2). It is followed by a presentation of hints on LaTeX (Section 4). The paper closes with a conclusion and an outlook on future work (Section 5).

2 Related Work

Winery [2] is a graphical modeling tool. The whole idea of TOSCA is explained by Binz et al. [1].

Tools for the literature search are, among others, ScienceOS, dblp, Google Scholar, IEEE Xplore, and the ACM Digital Library; JabRef manages the found papers, fetches the bibliographic data by DOI, and keeps the PDFs. If the search is systematic (as in a literature review), the keywords, the time span, and the inclusion criteria belong into the thesis so that a reader can repeat the search. Writing hints for student theses are given in the open-access book *Student Thesis Projects*[1]. More hints are given in *Writing for Computer Science*[2].

3 Writing in Markdown

This section uses plain Markdown, not LaTeX. Its source is the file `content-markdown.md`, which the main file includes with the `markdownInput` command of the `markdown` package[3]. This needs LuaLaTeX. Text can be *emphasized* or **bold**, and a citation looks like this [3]. Section 4.15 shows more Markdown.

4 LaTeX Hints

This section contains hints on writing LaTeX. It focuses on minimal examples, which can be directly adapted to the content

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short-conference-name, City, Country
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ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/9999997.9999999

[1] https://vdf.ch/product/student-thesis-projects-en.html
[2] https://doi.org/10.1007/978-1-4471-6639-9
[3] https://ctan.org/pkg/markdown

4.1 Handling of paragraphs

One sentence per line. This rule is important for the usage of version control systems. A new line is generated with a blank line. As you would do in Word: New paragraphs are generated by pressing enter. In LaTeX, this does not lead to a new paragraph as LaTeX joins subsequent lines. To start a new paragraph, press enter twice. This leads to an empty line. In word, there is the functionality to press shift and enter. This leads to a hard line break. The text starts at the beginning of a new line. In LaTeX, you can do that by using two backslashes (\\).
Running text seldom needs a hard line break.

Please do *not* use two backslashes for new paragraphs. For instance, this sentence belongs to the same paragraph, whereas the last one started a new one. Andrew Stacey gives a long motivation for that in the section "Line Wrapping" of his *Document Revision System*[4].

Corresponding LaTeX code of paper-basic-listings.tex

```
745 œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
746 One sentence per line.
747 This rule is important for the usage of version control
        ↪ systems.
748 A new line is generated with a blank line.
749 As you would do in Word:
750 New paragraphs are generated by pressing enter.
751 In LaTeX, this does not lead to a new paragraph as LaTeX
        ↪  joins subsequent lines.
752 To start a new paragraph, press enter twice.
753 This leads to an empty line.
754 In word, there is the functionality to press shift and
        ↪ enter.
755 This leads to a hard line break.
756 The text starts at the beginning of a new line.
757 In LaTeX, you can do that by using two backslashes (\
        ↪ textbackslash\textbackslash).
758 \\
759 Running text seldom needs a hard line break.
760
761 Please do \textit{not} use two backslashes for new
        ↪ paragraphs.
762 For instance, this sentence belongs to the same
        ↪ paragraph, whereas the last one started a new
        ↪ one.
763 Andrew Stacey gives a long motivation for that in the
        ↪ section \enquote{Line Wrapping} of his \href{
        ↪ https://loopspace.mathforge.org/HowDidIDoThat/
        ↪ TeX/VCS/#section.3}{\emph{Document Revision
        ↪ System}}\footnote{\url{https://loopspace.
        ↪ mathforge.org/HowDidIDoThat/TeX/VCS/}}.
```

Once the text is final, single paragraphs can be polished. If the last line of a paragraph holds only a word or two, or is nearly full, let TeX set the paragraph one line shorter or longer: write `\looseness=-1` (or `=1`) as the first thing in that paragraph. TeX resets `\looseness` at the end of every paragraph, so only that one paragraph is affected. With LuaLaTeX, a lone short word as the last line is avoided automatically (package impnattypo).

[4]https://loopspace.mathforge.org/HowDidIDoThat/TeX/VCS/

4.2 Notes separated from the text

The package mindflow enables writing down notes and annotations in a way so that they are separated from the main text.

This is a small note.

Corresponding LaTeX code of paper-basic-listings.tex

```
775 œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
776 \begin{mindflow}
777 This is a small note.
778 \end{mindflow}
```

4.3 Handling TODOs

Marked text.

Corresponding LaTeX code of paper-basic-listings.tex

```
783 œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
784 \textmarker{Marked text.}
```

With `\textmarker`, only the text color is changed, since this also works well for individual words.
Marked text.

Corresponding LaTeX code of paper-basic-listings.tex

```
789 œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
790 \textcomment{Marked text.}{A comment on it.}
```

Manual marking for text that has been changed since the last version.

Corresponding LaTeX code of paper-basic-listings.tex

```
793 œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
794 \modified{Manual marking for text that has been changed
        ↪ since the last version.}
```

This is some text. Changed text.

Corresponding LaTeX code of paper-basic-listings.tex

```
797 œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
798 This is some text.
799 \change{FL1: text adjusted}{Changed text}.
```

Here is a comment.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
Here is a comment\sidecomment{Comment}.
```



TODO!

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
\todo{A lot of text still needs to be produced here}
```

Several authors can mark their edits and comments with the phfcc package; the initials of the author appear once per page in the margin. Define one command per author in the preamble (`\phfMakeCommentingCommand`); `\phfDisableCommentingCommands` hides everything for the final version.

Quantum mechanics developed quickly thanks to the works of Schrödinger, Heisenberg, and Dirac, who drew inspiration from earlier ideas by Planck and Einstein [Einstein must not be missing here.] ~~This sentence is to be removed.~~ **!!! This is important. !!!**

FA

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
Quantum mechanics developed quickly thanks to the works
    ↪ of Schrödinger, Heisenberg, and Dirac,
who drew inspiration from earlier ideas by Planck \FA{
    ↪ and Einstein} \FA[Einstein must not be missing
    ↪ here.]
\FA*{This sentence is to be removed.} \FA!{This is
    ↪ important.}
```

4.4 Hyphenation

LaTeX automatically hyphenates words. When using microtype, there should be fewer hyphenations than in other settings. You might still need to tweak the hyphenation. Here are some hints:

In case you write “application-specific”, then the word will only be hyphenated at the dash. You can also insert `\allowbreak{}` into the word, as in application-specific, but this is much more effort.

You can now write words containing hyphens which are hyphenated at other places in the word. For that, write "= instead of -, as in application"=specific. This is enabled by an additional configuration of the babel package.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
In case you write \enquote{application-specific}, then
    ↪ the word will only be hyphenated at the dash.
You can also insert \verb1\allowbreak{}1 into the word,
    ↪ as in applica\allowbreak{}tion-specific, but
    ↪ this is much more effort.

You can now write words containing hyphens which are
    ↪ hyphenated at other places in the word.
For that, write \verb1"=1 instead of \verb1-1, as in
    ↪ application"=specific. % chktex 18
This is enabled by an additional configuration of the
    ↪ babel package.
```

4.5 Typesetting Units

Numbers can be written plain text (such as 100), by using the siunitx package as follows: 100 $\frac{\mathrm{km}}{\mathrm{h}}$, or by using plain LaTeX (and math mode): $100 \frac{km}{h}$.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
Numbers can be written plain text (such as 100), by
    ↪ using the \href{https://ctan.org/pkg/siunitx}{
    ↪ siunitx} package as follows:
\SI{100}{\km\per\hour},
or by using plain \LaTeX{} (and math mode):
$100 \frac{\mathit{km}}{h}$.
```

5 % of 10 kg

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
\SI{5}{\percent} of \SI{10}{kg}
```

Numbers are automatically grouped: 123 456.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
Numbers are automatically grouped: \num{123456}.
```

4.6 Surrounding Text by Quotes

Please use the “enquote command” to quote something. Quoting with “quote” or “quote” also works.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
Please use the \enquote{enquote command} to quote
    ↪ something.
Quoting with "`quote"' or ``quote'' also works. % chktex
    ↪ 18
```

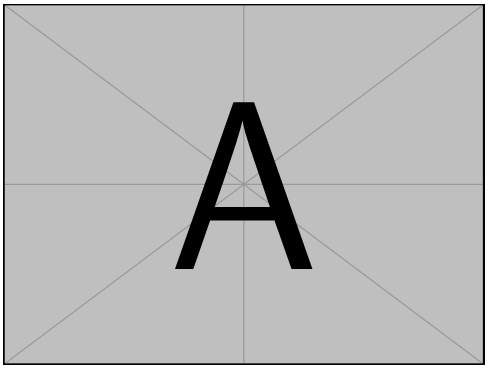


Figure 1: Example figure for zcref demo

Table 1: Example table for zcref demo

Heading1	Heading2
One	Two
Three	Four

4.7 Zref-clever examples

Demonstration of zref-clever: Zcref (short for zcref[S]) at the beginning of a sentence, zcref in all other cases.

Figure 1 shows a simple fact, although Figure 1 could also show something else.

Table 1 shows a simple fact, although Table 1 could also show something else.

Section 4.7 shows a simple fact, although Section 4.7 could also show something else.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
\Zcref{fig:ex:cref} shows a simple fact, although \zcref
      ↪ {fig:ex:cref} could also show something else.

\Zcref{tab:ex:cref} shows a simple fact, although \zcref
      ↪ {tab:ex:cref} could also show something else.

\Zcref{sec:ex:cref} shows a simple fact, although \zcref
      ↪ {sec:ex:cref} could also show something else.
```

4.8 Figures

Figure 2 shows something interesting.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
\Zcref{fig:label} shows something interesting.

\begin{figure}
  \centering
  \includegraphics[width=.8\linewidth]{example-image-
      ↪ golden}
  \caption[Simple Figure]{
    Simple Figure.
    Based on \citet{mwe}.
  }
  \label{fig:label}
\end{figure}
```

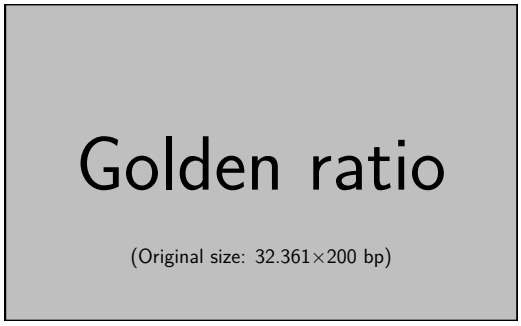


Figure 2: Simple Figure. Based on Scharrer [3].



Figure 3: Ulm Minster. Photo: wkrauth, CC0, via Wikimedia Commons.

Photos in JPEG or PNG format are included the same way, as shown in Figure 3.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
\begin{figure}
  \centering
  \includegraphics[width=.5\linewidth]{figures/ulm-
      ↪ minster.jpg}
  \caption{Ulm Minster. Photo: wkrauth, CC0, via
      ↪ Wikimedia Commons.}
  \label{fig:jpg}
\end{figure}
```

4.9 Sub Figures

An example of two sub figures is shown in Figure 4.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
\begin{figure}[!b]
  \centering
  \subfloat[Case I]{\includegraphics[width=.4\linewidth
      ↪ ]{example-image-a}%
    \label{fig:first_case}}
  \hfil
  \subfloat[Case II]{\includegraphics[width=.4\linewidth
      ↪ ]{example-image-b}%
    \label{fig:second_case}}
  \caption{Example figure with two sub figures.}
  \label{fig:two_sub_figures}
\end{figure}
```

4.10 Figures with tikz

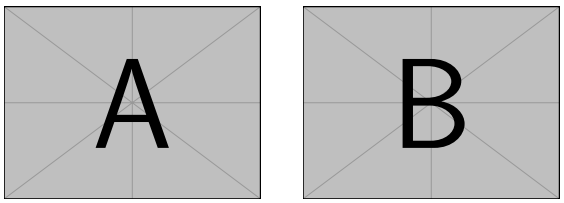
The tikz package creates graphics programmatically. This package generates grids and other regular structures.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
\begin{figure}[ht]
  \centering
  \begin{tikzpicture}
    \draw(0,0) rectangle (4,4);
    \foreach \x in {0.5,1,1.5,2,2.5,3,3.5}
    \foreach \y in {0.5,1,1.5,2,2.5,3,3.5}
    \draw(\x,\y) circle (1pt);
  \end{tikzpicture}
  \caption{A regular grid generated easily with two for
      ↪ loops.}\label{fig:tikz_example}
\end{figure}
```

4.11 Plots with pgfplots

The pgfplots package plots functions directly in LaTeX, similar to MATLAB or gnuplot. Some visual examples are available in the package documentation[5].



(a) Case I (b) Case II

Figure 4: Example figure with two sub figures.

[5]http://texdoc.net/pkg/pgfplots

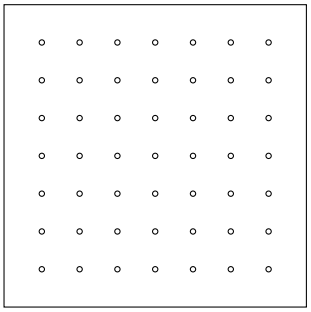


Figure 5: A regular grid generated easily with two for loops.

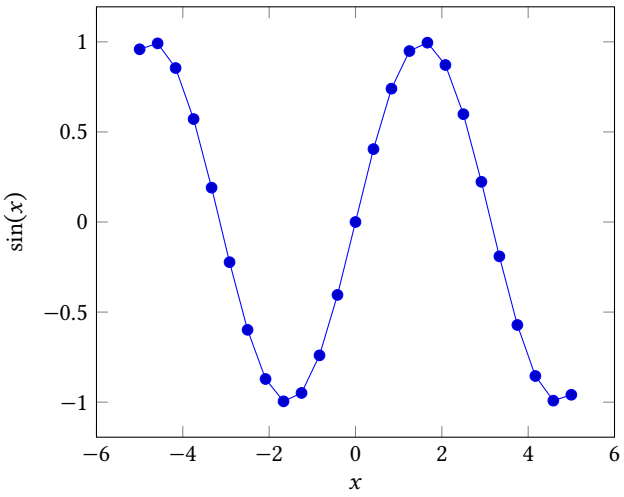


Figure 6: Plot of $\sin(x)$ directly inside the figure environment with pgfplots.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
\begin{figure}[ht]
  \centering
  \begin{tikzpicture}
    \begin{axis}[xlabel=$x$, ylabel=$\sin(x)$]
      \addplot {sin(deg(x))};  % Plot the sine function
    \end{axis}
  \end{tikzpicture}
  \caption{Plot of $\sin(x)$ directly inside the figure
      ↪ environment with pgfplots.}
  \label{fig:pgfplots_example}
\end{figure}
```

Coordinates can also be given explicitly instead of as a function:

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
\begin{figure}[ht]
```

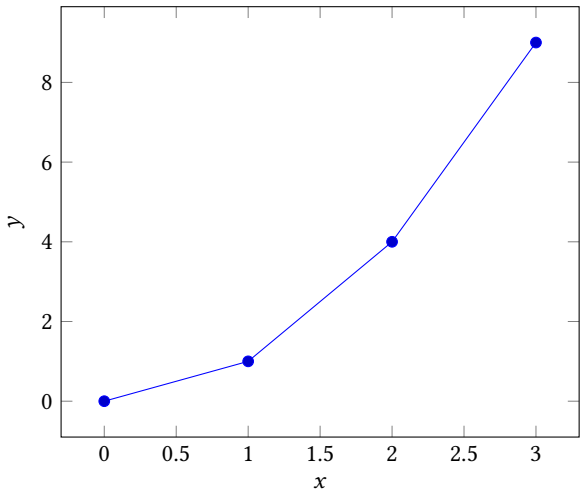


Figure 7: Explicit coordinates plotted with pgfplots.

Table 2: Simple Table

Heading1	Heading2
One	Two
Three	Four

```
  \centering
  \begin{tikzpicture}
    \begin{axis}[xlabel=$x$, ylabel=$y$]
      \addplot coordinates {(0,0) (1,1) (2,4) (3,9)};  %
          ↪ Plot explicit coordinates
    \end{axis}
  \end{tikzpicture}
  \caption{Explicit coordinates plotted with pgfplots.}
  \label{fig:pgfplots_coords}
\end{figure}
```

4.12 Tables

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
\begin{table}
  \caption{Simple Table}
  \label{tab:simple}
  \centering
  \begin{tabular}{ll}
    \toprule
    Heading1 & Heading2 \\
    \midrule
    One      & Two      \\
    Three    & Four     \\
    \bottomrule
  \end{tabular}
\end{table}
```

A table caption goes above the table (`\caption` before the `tabular`), a figure caption below the figure.

Table 3: Table with colored cells

Test	Result
A	pass
B	fail

Rows and cells can be colored with `\rowcolor` and `\cellcolor` from the colortbl package, which xcolor loads with its option `table`.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
\begin{table}
  \caption{Table with colored cells}
  \label{tab:colored}
  \centering
  \begin{tabular}{ll}
    \toprule
    \rowcolor{gray!20}
    Test & Result                    \\
    \midrule
    A    & \cellcolor{green!20}pass \\
    B    & \cellcolor{red!20}fail   \\
    \bottomrule
  \end{tabular}
\end{table}
```

Table 4: Table with a footnote

Heading1	Heading2[6]
One	Two

A `\footnote` inside a floating table is lost. Use `\tablefootnote` from the tablefootnote package instead: it is set as a normal footnote at the bottom of the page. If the notes should appear directly below the table instead (common in journals), use the threeparttable package.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
\begin{table}
  \caption{Table with a footnote}
  \label{tab:footnote}
  \centering
  \begin{tabular}{ll}
    \toprule
    Heading1 & Heading2\tablefootnote{A footnote inside
        ↪ a table.} \\
    \midrule
    One      & Two
        ↪
        ↪ \\
    \bottomrule
```

[6] A footnote inside a table.

Table 5: Wide table scaled to the line width

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1.11	2.22	3.33	4.44	5.55	6.66	7.77	8.88	9.99	10.10	11.11	12.12
Rain	Snow	Wind	Sun	Fog	Hail	Heat	Storm	Mist	Frost	Sleet	Calm

Table 6: Table with diagonal line

Row \ Column	A	B
1	x	y

```
  \end{tabular}
\end{table}
```

A table wider than the text runs into the margin. `\begin{adjustbox}{max width=\linewidth}` from the adjustbox package scales it down to the line width (the column width in a two-column layout), and only if it is too wide. Scaling shrinks the font, so for a table with long text in its cells, prefer wrapping the text with the column type `X` of the tabularx package, and for a very wide table, a landscape page (see `pdflscape`).

Corresponding LaTeX code of `paper-basic-listings.tex`

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
\begin{table}
  \caption{Wide table scaled to the line width}
  \label{tab:wide}
  \centering
  \begin{adjustbox}{max width=\linewidth}
    \begin{tabular}{lllllllllll}
      \toprule
      Jan  & Feb  & Mar  & Apr  & May  & Jun  & Jul  &
          ↪ Aug  & Sep  & Oct  & Nov  & Dec  \\
      \midrule
      1.11 & 2.22 & 3.33 & 4.44 & 5.55 & 6.66 & 7.77 & 8.
          ↪ 88  & 9.99 & 10.10 & 11.11 & 12.12 \\
      Rain & Snow & Wind & Sun  & Fog  & Hail & Heat &
          ↪ Storm & Mist & Frost & Sleet & Calm  \\
      \bottomrule
    \end{tabular}
  \end{adjustbox}
\end{table}
```

Corresponding LaTeX code of `paper-basic-listings.tex`

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
% Adapted from https://tex.stackexchange.com/a
    ↪ /468994/9075
\begin{table}
  \caption{Table with diagonal line}
  \label{tab:diag}
  \centering
  \begin{tabular}{|l|c|c|} % chktex 44
    \hline % chktex 44
    \diagbox{Row}{Column} & A & B \\
    \hline % chktex 44
    1                     & x & y \\
    \hline % chktex 44
  \end{tabular}
\end{table}
```

```
<listing name="example">
  Floating
</listing>
```

Listing 2: Example XML listing – placed as floating figure

4.13 Source Code

Listing 1 shows source code written in XML. Line 2 contains a comment.

```
<listing name="example">
  <!-- comment -->
  <content>not interesting</content>
</listing>
```

Listing 1: Example XML Listing

Corresponding LaTeX code of `paper-basic-listings.tex`

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
% After a period, LaTeX sets the wider sentence space by
    ↪  default. The exception: a period after a
    ↪ capital letter
% is taken for an abbreviation or initial ("T. Scientist
    ↪ ", "NASA") and gets the narrow inter-word space.
% "XML." ends a sentence, so \@ before the period
    ↪ restores the sentence space.
% (The other way round, "e.g.\ details" forces the
    ↪ narrow space after a lowercase abbreviation; ~
    ↪ is a
% non-breaking inter-word space, as in "Prof.~Dr.~Eike
    ↪ Kraft".)
% An abbreviation set with \gls{xml} needs no \@:
    ↪ glossaries ends it with a non-letter, so the
    ↪ period after it
% gets the sentence space. The plain "XML" here is for
    ↪ documents without the glossaries package.
\Zcref{lst:XML} shows source code written in XML\@.
\Zcref{line:comment} contains a comment.

\begin{lstlisting}[
  language=XML,
  caption={Example XML Listing},
  label={lst:XML}]
<listing name="example">
  <!-- comment --> (* \label{line:comment} *)
  <content>not interesting</content>
</listing>
\end{lstlisting}
```

One can also add `float` as parameter to have the listing floating. Listing 2 shows the floating listing.

```
{
  key: "value"
}
```

Listing 3: Example JSON listing – placed as floating figure

```
public class Hello {
    public static void main (String[] args) {
        System.out.println("Hello World!");
    }
}
```

Listing 4: Example Java listing

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
\begin{lstlisting}[
  % one can adjust spacing here if required
  % aboveskip=2.5\baselineskip,
  % belowskip=-.8\baselineskip,
  float,
  language=XML,
  caption={Example XML listing -- placed as floating
       ↪ figure},
  label={lst:flXML}]
<listing name="example">
  Floating
</listing>
\end{lstlisting}
```

One can also typeset JSON as shown in Listing 3.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
\begin{lstlisting}[
  float,
  language=json,
  caption={Example JSON listing -- placed as floating
       ↪ figure},
  label={lst:json}]
{
  key: "value"
}
\end{lstlisting}
```

Java is also possible as shown in Listing 4.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
\begin{lstlisting}[
  caption={Example Java listing},
  label=lst:java,
  language=Java,
  float]
public class Hello {
    public static void main (String[] args) {
        System.out.println("Hello World!");
    }
}
\end{lstlisting}
```

4.14 Itemization

One can list items as follows:

- Item One
- Item Two

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
\begin{itemize}
  \item Item One
  \item Item Two
\end{itemize}
```

With the package paralist, one can create itemizations with lesser spacing:

- Item One
- Item Two

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
\begin{compactitem}
  \item Item One
  \item Item Two
\end{compactitem}
```

One can enumerate items as follows:

(1) Item One
(2) Item Two

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
\begin{enumerate}
  \item Item One
  \item Item Two
\end{enumerate}
```

With the package paralist, one can create enumerations with lesser spacing:

(1) Item One
(2) Item Two

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…——„“”,‘’
\begin{compactenum}
  \item Item One
  \item Item Two
```

Right	Left	Centered
12	12	12
123	123	123

Table 7: A table written in Markdown

```
\end{compactenum}
```

With paralist, one can even have all items typeset after each other and have them clean in the TeX document:

(1) The first item, (2) the second item, and (3) the third item appear in one line.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—––„“”,‘’
\begin{inparaenum}
  \item The first item,
  \item the second item, and
  \item the third item appear in one line.
\end{inparaenum}
```

4.15 Writing in Markdown

If you are new to LaTeX, you can start by writing your text in Markdown using the markdown package. LaTeX then takes care of the typesetting. Markdown and LaTeX can be mixed: Table 7 is written in Markdown and referenced from LaTeX. A whole section can be written in a separate Markdown file included by `\markdownInput`, as Section 3 shows. The markdown-latex-quickstart shows a complete document written in Markdown.

Text can be *emphasized* or **bold**, and code is set in a monospaced font. A citation looks like this [3], and here is a footnote.[7] Referenced footnotes work as well[8].

- first point
- second point
  - sub-point

1. firstly
2. secondly

Table 7 shows a table with caption. Its identifier `{#tbl:markdown}` is referenced by `<#tbl:markdown>`.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—––„“”,‘’
  \begin{markdown}
Text can be *emphasized* or **bold**, and `code` is set
      ↪ in a monospaced font.
A citation looks like this [@mwe], and here is a
      ↪ footnote.^[This is an inline footnote.]
Referenced footnotes work as well[^detail].

[^detail]: This is a footnote defined at a different
      ↪ place in the text.

- first point
- second point
  - sub-point

1) firstly
2) secondly

| Right | Left | Centered |
|------:|:-----|:--------:|
|    12 | 12   |    12    |
|   123 | 123  |   123    |

: A table written in Markdown {#tbl:markdown}

<#tbl:markdown> shows a table with caption.
Its identifier `{#tbl:markdown}` is referenced by `<#tbl:
      ↪ markdown>`.
\end{markdown}
```

4.16 Other Features

The words “workflow” and “dwarflike” can be copied from the PDF and pasted to a text file.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—––„“”,‘’
The words \enquote{workflow} and \enquote{dwarflike} can
      ↪  be copied from the PDF and pasted to a text
      ↪ file.
```

The symbol for powerset is now correct: $\wp$ and not a Weierstrass p ($\wp$).

$\wp(1,2,3)$

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—––„“”,‘’
The symbol for powerset is now correct: $\powerset$ and
      ↪ not a Weierstrass p ($\wp$).

$\powerset({1,2,3})$
```

Brackets work as designed: <test> One can also input backticks in verbatim text: `` `test` ``.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—––„“”,‘’
Brackets work as designed:
<test>
One can also input backticks in verbatim text: \verb|`
      ↪ test`|.
```

Acronyms can be typeset in running text with `\initialism`,

[7] This is an inline footnote.
[8] This is a footnote defined at a different place in the text.

which sets them as small caps that blend into the surrounding lowercase text. It is the lightweight alternative to the full acronym and glossary management (`\gls`, with an automatic list of abbreviations) provided by the thesis variants: `\initialism` only typesets the acronym and does not add it to any list. The macro and the ready-made acronyms (`\OMG`, `\BPEL`, `\BPMN`, `\UML`) are defined in `commands.tex`, where you can add your own.

The UML is maintained by the OMG; related standards include BPEL and BPMN. Any acronym can be set inline with REST or HTTP.

Corresponding LaTeX code of `paper-basic-listings.tex`

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…—–„“”,‘’
The \UML is maintained by the \OMG; related standards
    ↪ include \BPEL and \BPMN.
Any acronym can be set inline with \initialism{REST} or
    ↪ \initialism{HTTP}.
```

5 Conclusion and Outlook

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

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Acknowledgments

Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work should be included in an acknowledgment section, which comes right before the reference section in your document [4].

For more LaTeX hints for ACM read on at https://www.acm.org/publications/taps/latex-best-practices.

References

[1] Tobias Binz, Gerd Breiter, Frank Leymann, and Thomas Spatzier. 2012. Portable Cloud Services Using TOSCA. *IEEE Internet Computing* 16, 03 (May 2012), 80–85. doi:10.1109/mic.2012.43

[2] Oliver Kopp et al. 2013. Winery – A Modeling Tool for TOSCA-based Cloud Applications. In *Proceedings of 11th International Conference on Service-Oriented Computing (ICSOC'13) (LNCS, Vol. 8274)*. Springer Berlin Heidelberg, 700–704. doi:10.1007/978-3-642-45005-1_64

[3] Martin Scharrer. 2017. *The* mwe *Package*. http://texdoc.net/mwe

[4] Boris Veytsman. 2021. LaTeX Class for the Association for Computing Machinery – Acknowledgement Information. Retrieved September 17, 2021 from https://github.com/borisveytsman/acmart/blob/1704c8bf7eee92a1515ff755f5118b6a22bb1f8e/samples/samples.dtx#L709

All links were last followed on October 5, 2020.